ASSIGNMENT: 1
Malware Analysis: CSCF3203

SCENARIO: Incident Situation Report: Critical Infrastructure Compromise
Date: 8th Feb, 2026
Severity Level: Critical **Incident Type**: Advanced Persistent Threat / Human-Operated Ransomware

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Executive Summary

The Computer Security Incident Response Team (CSIRT) received a suspicious executable sample as part of an investigation of a compromised unstated critical infrastructure. The incident has been categorized as an Advanced Persistent Threat (APT) to the infrastructure involving human-operated ransomware activity at multiple levels.

The Malware Analysis Team received the suspicious file in a password-protected format in accordance with malware handling rules and regulations. Upon extraction and primary inspection, the file was identified as a 32-bit Microsoft .NET (MSIL) executable, seeming as a legitimate setup utility. Static and dynamic analysis of the sample confirmed that the sample exhibits malicious characteristics and aligns with behaviors commonly associated with Remote Access Trojans (RATs) and staged intrusion frameworks.

The executable demonstrates extensive system reconnaissance activity, including registry enumeration, environment profiling, boot directory inspection, and architecture detection. The high entropy value of the binary suggests the presence of encrypted or obfuscated contents of the executable.

Threat intelligence via VirusTotal antivirus scanning platform resulted in 53 out of 72 security vendors classifying the file as malicious, with multiple detections referencing MSIL-based trojans and RAT families.

Based on the conducted analysis, the sample is confirmed to be malicious and is assessed to represent an initial access or persistence component potentially deployed before ransomware execution within the compromised infrastructure.

1. Sample Information

The suspicious sample was extracted from a password-protected .zip file and analyzed in a network-isolated sandboxed environment. The file initially had no extension and was identified using hash-based naming conventions.

1.1 File Identification

- File Name: setup.exe

- Analyzed File Name (Original): f91b8fc07602d0d7c8639a7554bbcec478187c259f7441668beb6966421cf8cc
- File Type: PE32 Executable (GUI)
- Architecture: 32-bit (x86)
- Platform: Microsoft Windows
- SubSystem: Win32 GUI
- Framework: .NET Framework v2.0.50727 (MSIL Assembly)
- File Size: 456,192 bytes (445.50 KB)
- Entropy: 5.64 (Low)

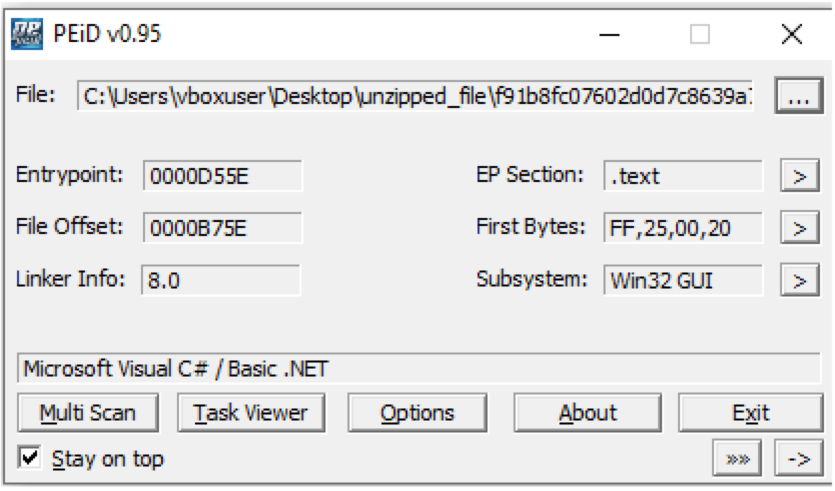


Fig. 1.1 Overview of the sample provided by PEiD v0.95

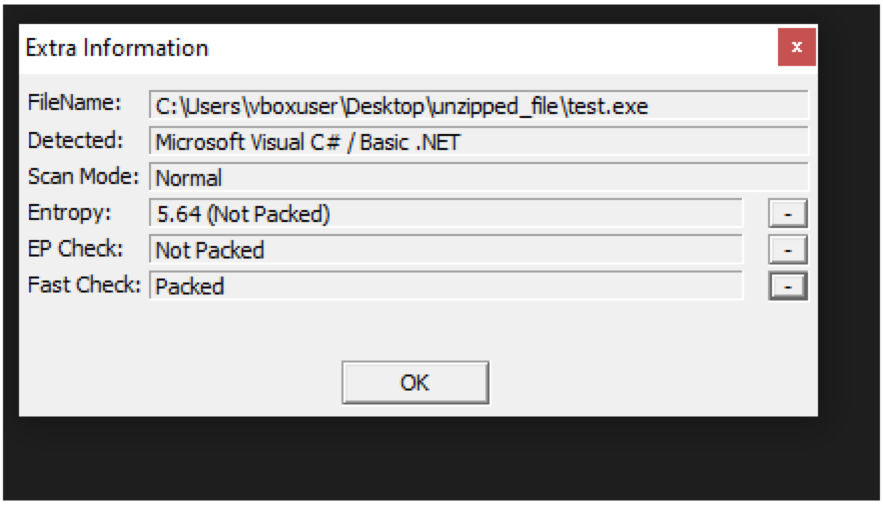


Fig. 1.2 Obfuscation nature of the sample

2.2 Cryptographic Hashes

MD5 - d65f0eac61b375293969dd1398fab2b5

SHA256 - f91b8fc07602d0d7c8639a7554bbcec478187c259f7441668beb6966421cf8cc

```
C:\Users\vboxuser\Desktop\unzipped_file>certutil -hashfile f91b8fc07602d0d7c8639a7554bbcec478187c259f7441668beb6966421cf
8cc MD5
MD5 hash of f91b8fc07602d0d7c8639a7554bbcec478187c259f7441668beb6966421cf8cc:
d65f0eac61b375293969dd1398fab2b5
CertUtil: -hashfile command completed successfully.
```

Fig. 2.2 MD5 Hash of file generated using certutil function of windows

```
:\Users\vboxuser\Desktop\unzipped_file>certutil -hashfile f91b8fc07602d0d7c8639a7554bbcec478187c259f7441668beb6966421cf
3cc SHA256
SHA256 hash of f91b8fc07602d0d7c8639a7554bbcec478187c259f7441668beb6966421cf8cc:
f91b8fc07602d0d7c8639a7554bbcec478187c259f7441668beb6966421cf8cc
CertUtil: -hashfile command completed successfully.
```

Fig 2.3 SHA256 Hash of file generated using certutil function of windows

2.3 Compilation Metadata

- Compiler: Microsoft Linker 8.0
- Compilation Timestamp**:** Sun Mar 20 15:28:53 2011 (UTC)
- Language: .NET (MSIL / C# or VB.NET)
- Signature: Not Signed

The compilation timestamp significantly predates the file creation timestamp observed in the lab environment, suggesting potential timestamp manipulation or reuse of older build configurations.

2.4 Threat Intelligence Correlation

Multi-engine antivirus scanning results indicate:

- Detection Rate: 53 / 72 security vendors flagged as malicious
- Threat Classification: Trojan (MSIL-based)
- Associated Family Labels: MSIL, msilperseus, DarkComet (various vendors)

This level of detection strongly supports malicious classification and aligns with the observed behavioral indicators during analysis.

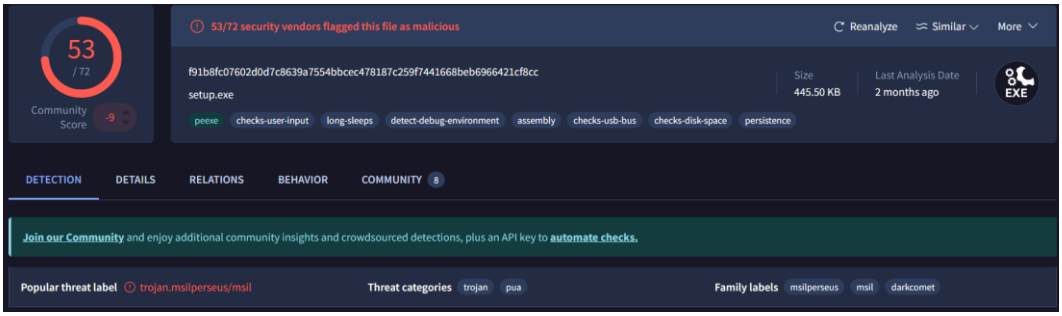


Fig. 2.4 Virustotal Detection Overview

2. Sample Acquisition & Analysis Environment

3.1 Sample Acquisition Procedure

The sample (suspicious file) was received from the Computer Security Incident Response Team (CSIRT) in a password-protected compressed archive format (.zip). This handling method was used to prevent accidental execution of the suspect file and infecting the handling machine.

Upon receiving the suspect file:

- The archive was inspected in a network-isolated environment.
- The default Windows extraction utility failed to extract the contents and showed an error with the extraction of files without an extension
- The archive was successfully extracted using 7-Zip, a free and open source file archiver with a high compression ratio
- The file, upon extraction, had no file extension and was named using a hash-like identifier.
- The absence of a file extension and the hash-based filename format indicate deliberate obfuscation intended to avoid generic identification.

Before Execution:

- Cryptographic hash values were calculated.
- The file structure was verified to confirm it was a valid Portable Executable (PE).
- The file was renamed with a `.exe` extension, only for analysis via set rules and tools.

3.2 Analysis Environment

All analysis was conducted in an isolated virtualized operating system environment to ensure containment and prevent unintended system compromise.

The following environments were used:

- Windows 10 (64-bit)
- Windows 8
- FLARE VM

The systems were configured with:

- Process monitoring tools
- Registry monitoring
- Network activity monitoring (to monitor command-and-control (C2) communication attempts)
- Static and reverse engineering tools

3.3 Analysis Tools

- PEiD
- DnSpy
- PEStudio
- FLOSS (FireEye Labs Obfuscated String Solver)
- UPX (packing detection)

- ILSpy
- DIE (Detect It Easy)

Dynamic Analysis Tools:

- Process Monitor (ProcMon)
- DNS monitoring utilities
- fakenet

Each tool was used to validate findings both the static and dynamic nature of the sample file.

3. Static Analysis

Static analysis was conducted on the received sample file to examine the internal structure of the executable without its file execution. The objective was to determine file characteristics, packing status, embedded resources, and potential malicious indicators.

4.1 File Identification Type

The file had no file extension, and after verifying the Portable Executable signature, the file was renamed with a .exe extension for analysis using the above-specified tools.

PEiD was used to confirm the file format. The results confirmed PE32 executable with Intel 80386 architecture of CLR Version 2.0.50727. The file was a .NET file written in either C# / Basic .NET.

DnSpy:

The entry point is ns2.Class11.Main

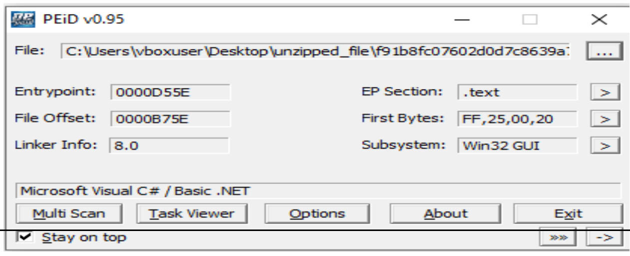
```
// Entry point: ns2.Class11.Main
// Timestamp: 4D861D35 (3/20/2011 8:58:53 PM)
```

Fig 4.1.1

As it is a .NET executable execution is done by the Common Language Runtime (CLR), meaning the primary logic is implemented in managed Intermediate Language (IL) rather than native x86 set of predefined instructions.

File type	Win32 EXE executable windows win32 pe peexe
Magic	PE32 executable (GUI) Intel 80386 Mono/.Net assembly, for MS Windows
TrID	Generic CIL Executable (.NET, Mono, etc.) (66.5%) | Win64 Executable (generic) (9.5%) | Win32 Dynamic Link Library (generic) (5.9%) | Win16 NE executable (generic) (4.5...
DetectItEasy	PE32 | Library: .NET (v2.0.50727) | Linker: Microsoft Linker (8.0)
Magika	PEBIN
File size	445.50 KB (456192 bytes)
PEiD packer	Microsoft Visual C# / Basic .NET

Fig 4.1.2



4.2 Packing and Obfuscation Details

Two primary tools were used to detect whether the executable was packed or unpacked. And upon opening the file in DnSpy to analyse the source code, it was not obfuscated either thus validating the results given by the tools. The tools were:

- UPX
- PEiD
- DnSpy

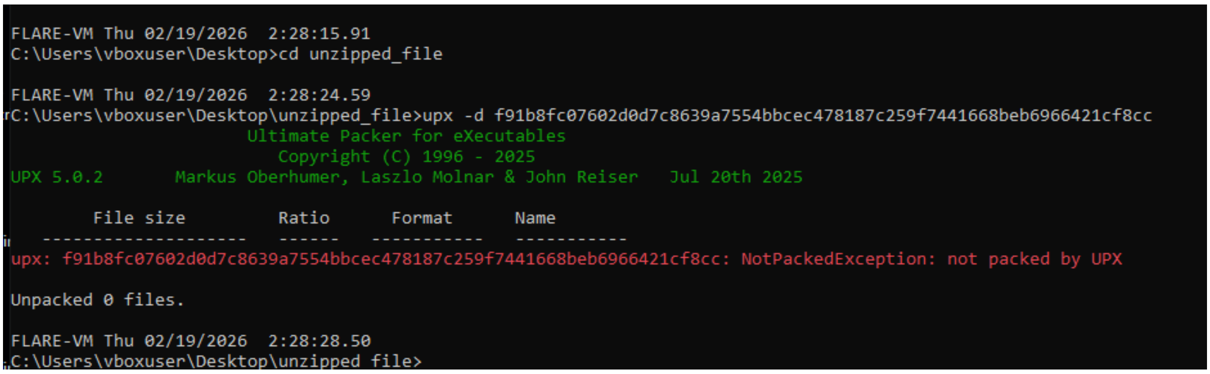


Fig 4.2.1: UPX analysis returned: “NOT PACKED BY UPX”

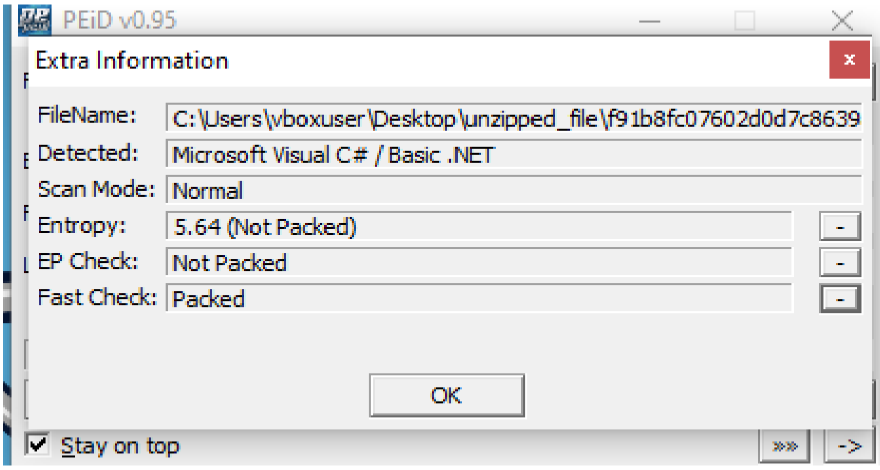


Fig 4.2.2: PEiD returns an Entropy of 5.64 so the sample is not packed

This helped in understanding that the file is not packed and not obfuscated.

4.3 File Properties

The sample file exhibits the properties as shown in the figure below:

Fig 4.3.1: File Properties using DIE tool

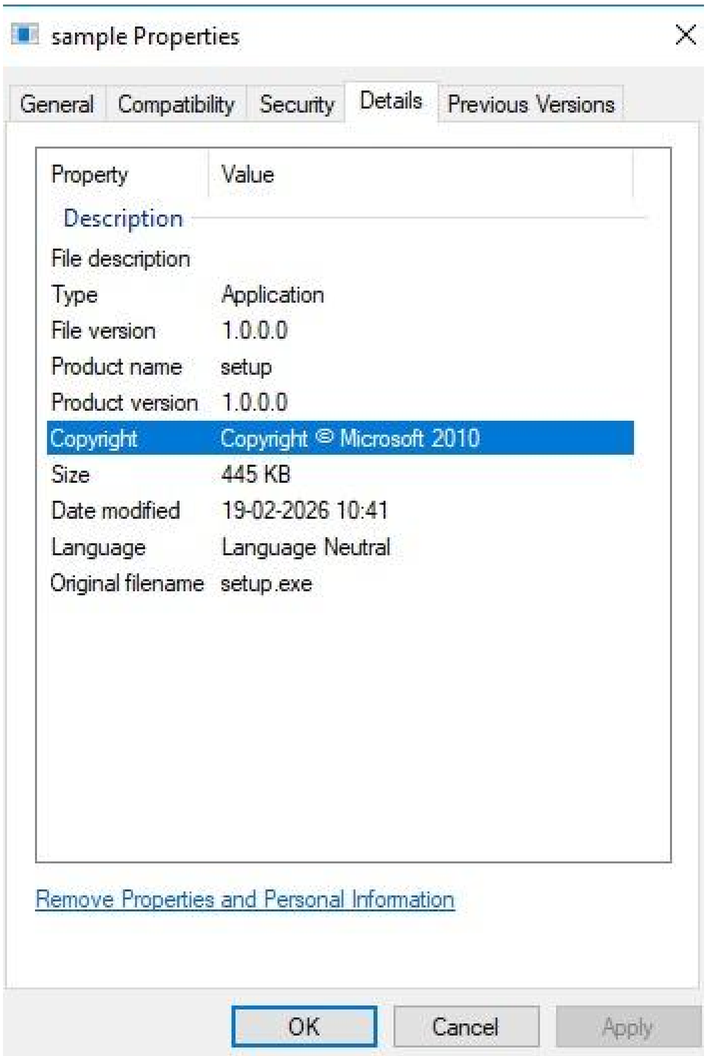


Fig. 4.3.2: File Properties using the Windows "Properties Function"

4.4 Import Library Analysis

Analysis of imported libraries revealed usage of:

- mscoree.dll (CLR execution)
- mscorlib.dll (.NET base library)
- advapi32.dll (Registry and security functions internal windows api)
- winsock32.dll (Network communication capability)
- combase.dll (COM interactions)
- SHCore.dll
- Winspool.drv
- kernal32
- fusion

The presence of `advapi32.dll` confirms the capability of the file to manipulate registry entries, which is later confirmed during Dynamic Analysis that it makes changes within different registry folders to create persistence. The inclusion of `winsock32.dll` indicates network capabilities, which could be us. `mscoree.dll` is the Runtime Execution Engine used by .NET applications to properly run, confirming execution within the .NET runtime environment. The presence of `kernal32.dll` displays the malware being able to use functions to interact with the OS to perform fundamentals like process creation, memory allocation, DLL loading etc. The rest libraries are used for proper execution of .NET files and processes.

Although import analysis alone does not confirm malicious activity, the combination of registry, networking, and managed execution libraries aligns with Trojan or RAT functionality.

Files accessed during trace:

By Path | By Folder | By Extension

Name	File Time	Total Events	Opens	Closes	Reads	Writes	Read B...	Write B...	Get ACL	Set ACL	Other
⊟ C:	5.9248855	1535	429	429	248	0	6342144	0	0	0	429
$Directory	0.0822957	9	0	0	9	0	36864	0	0	0	0
⊟ Users	0.0004839	1	0	0	1	0	512	0	0	0	0
⊟ mahes	0.0004839	1	0	0	1	0	512	0	0	0	0
⊟ Desktop	0.0004839	1	0	0	1	0	512	0	0	0	0
sample.exe	0.0004839	1	0	0	1	0	512	0	0	0	0
⊟ Windows	5.8421059	1525	429	429	238	0	6304768	0	0	0	429
⊟ SysWOW64	5.7889277	1468	429	429	181	0	5581312	0	0	0	429
AcGenral.dll	0.0044944	7	0	0	7	0	229376	0	0	0	0
AcLayers.dll	0.0014711	3	0	0	3	0	98304	0	0	0	0
KernelBase.dll	2.1528959	53	0	0	53	0	1637376	0	0	0	0
SHCore.dll	0.0073568	6	0	0	6	0	180224	0	0	0	0
advapi32.dll	0.0076700	6	0	0	6	0	196608	0	0	0	0
apphelp.dll	0.0065044	5	0	0	5	0	163840	0	0	0	0
⊟ boot	0.0938937	1287	429	429	0	0	0	0	0	0	429
mtldr32.exe	0.0000000	429	0	0	0	0	0	0	0	0	429
combase.dll	0.0076831	6	0	0	6	0	196608	0	0	0	0
kernel32.dll	0.0233264	22	0	0	22	0	696320	0	0	0	0
msacm32.dll	0.0007101	1	0	0	1	0	32768	0	0	0	0
mscoree.dll	0.0045176	6	0	0	6	0	196608	0	0	0	0
msctf.dll	0.0017702	2	0	0	2	0	65536	0	0	0	0
msvcrt.dll	0.1340849	1	0	0	1	0	32768	0	0	0	0
ntdll.dll	0.0645840	30	0	0	30	0	794624	0	0	0	0
propsys.dll	0.0138776	9	0	0	9	0	294912	0	0	0	0
shell32.dll	0.8240305	7	0	0	7	0	229376	0	0	0	0
ucrtbase.dll	0.0086759	5	0	0	5	0	163840	0	0	0	0
windows.storage.dll	2.4284280	7	0	0	7	0	229376	0	0	0	0
wininet.dll	0.0015436	2	0	0	2	0	65536	0	0	0	0
winspool.drv	0.0008423	2	0	0	2	0	65536	0	0	0	0
wsock32.dll	0.0005672	1	0	0	1	0	11776	0	0	0	0
⊟ System32	0.0531782	57	0	0	57	0	723456	0	0	0	0
ntdll.dll	0.0069177	7	0	0	7	0	110592	0	0	0	0
wow64.dll	0.0419211	45	0	0	45	0	610304	0	0	0	0
wow64cpu.dll	0.0043394	5	0	0	5	0	2560	0	0	0	0

Fig. 4.4: DLLs accessed during program execution

4.5 String Analysis (FLOSS)

FLOSS was used to recover obfuscated and decoded strings from thePres executable.

Results revealed:

- Large volume of human-readable and non-human-readable strings
- Function References and System-Related Identifiers
- No immediate plaintext cryptographic keys

```
Struct9
<Module>
refIID
8SFLhV3TDVdAPqjLLW.vFCXRvIkorVLyJYGcH
System.IO
set_IV
value__
System.Web
mscorlib
System.Collections.Generic
PropertyId
pcbRead
RijndaelManaged
cbReserved
pvReserved
dwReserved
Synchronized
Guid
ResolveMethod
method
Replace
FindResource
set_Mode
FileMode
CryptoStreamMode
CompressionMode
CipherMode
get_Unicode
a5b7ee8e-cbdf-4eff-9144-efd0c433f3fe
ppAsmCache
CreateAssemblyCache
EndInvoke
BeginInvoke
IDisposable
RuntimeFieldHandle
RuntimeTypeHandle
CloseHandle
GetTypeFromHandle
File
get_MainModule
ProcessModule
get_ManifestModule
get_FileName
ppName
GetName
pccDisplayName
szDisplayName
pszAssemblyName
pszName
pwzName
ComInterfaceType
ValueType
ResolveType
```

Figure 4.5.1 – Cryptographic and Encoding Related Strings Extracted Using FLOSS

The output shows encoding and encryption related strings such as `FromBase64String`, `Rijndael`, and `CryptoStream`. This indicates that the executable may perform encryption or data transformation during execution.

```
ToChar
BinaryReader
MD5CryptoServiceProvider
RSACryptoServiceProvider
DESCryptoServiceProvider
lpcwBuffer
```

Figure 4.5.2 – Cryptographic Provider and Cipher Configuration Strings

The presence of `MD5CryptoServiceProvider`, `RSACryptoServiceProvider`, and `CipherMode` suggests a structured cryptographic implementation. This indicates support for multiple encryption or hashing algorithms within the executable.

```
byte_0
string_0
ulong_0
asyncCallback_0
bool_0
get_CultureInfo_0
set_CultureInfo_0
cultureInfo_0
get_ResourceManager_0
resourceManager_0
intptr_0
object_0
iasyncResult_0
uint_0
short_0
sortedList_0
dictionary_0
Interface0
Delegate0
Attribute0
Stream0
Enum0
Class0
Struct0
smethod_11
Delegate11
Class11
Struct11
smethod_21
Class21
struct0_1
struct1_1
delegate12_1
struct2_1
struct3_1
class4_1
struct5_1
struct9_1
imethod_1
smethod_1
byte_1
string_1
long_1
bool_1
intptr_1
object_1
VirtualProtect_1
uint_1
short_1
Class25`1
Interface1
Delegate1
Class1
```

Figure 4.5.3 – Obfuscated Class and Method Identifiers Observed in the Executable

Generic names like `Class1`, `method_12`, and `string_0` indicate possible code obfuscation. This technique is commonly used to reduce readability and make static analysis more difficult.

```
libNewSize
dwMaxSize
grfStatFlag
Encoding
FromBase64String
ToBase64String
ToString
GetString
pstatstg
ComputeHash
Math
pszManifestFilePath
get_Length
pdwVersionHi
AsyncCallback
callback
FlushFinalBlock
TransformFinalBlock
pAsmBindSink
Marshal
Rijndael
System.ComponentModel
kernel32.dll
GetManifestResourceStream
FileStream
get_BaseStream
DeflateStream
CryptoStream
ppStream
MemoryStream
ppAsmItem
System
SymmetricAlgorithm
HashAlgorithm
Trim
ICryptoTransform
ppstm
Enum
get_MetadataToken
GetPublicKeyToken
pcbWritten
Main
dwOrigin
Join
System.IO.Compression
fusion
get_Location
System.Configuration
System.Globalization
System.Reflection
get_Position
set_Position
plibNewPosition
```

Figure 4.5.4 – System API and File Handling References Extracted by FLOSS

Strings such as `kernel32.dll`, `System.IO`, and `FileStream` show interaction with Windows APIs and file operations. This suggests that the program is accessing system resources to perform file manipulation and other actions to maintain persistence in the system.

5. Dynamic Analysis

The aim of dynamic analysis was to monitor the runtime behavior of the malware sample. The malware sample was analyzed to determine the system changes, registry interactions, file system interactions, persistence techniques, network connections and possible malicious functionality exhibited by the executable file.

5.1 Environment setup

The malware sample was run in an isolated virtual environment to avoid any unintended system modifications. A snapshot of the virtual environment was taken before the malware sample was run to restore the system state after the analysis.

Operating System: Windows 10(Virtual Machine)

Execution Environment: Isolated sandbox environment

Monitoring Environment: Process and system monitoring

The malware sample that was analyzed was named Setup.exe and was classified as a DaRCoMet Remote Access Trojan (RAT).

5.2 Tools Used

Two primary tools were used ProcMon(Process Monitor) and Process Explorer to monitor what processes were created, what actions were performed and process termination. Fakenet was used to impersonate a real network while the system was isolated via network adapters to observe any network connections made.

5.3 Behavioral Analysis

5.4.1 Process Execution

Upon execution, the malware created a new process under the name *sample.exe* and began interacting extensively with the Windows operating system. The process performed multiple system profiling operations and continuously accessed system resources.

No additional child processes were observed during initial execution.

5.4.2 Registry Activity

The malware performed extensive registry operations including:

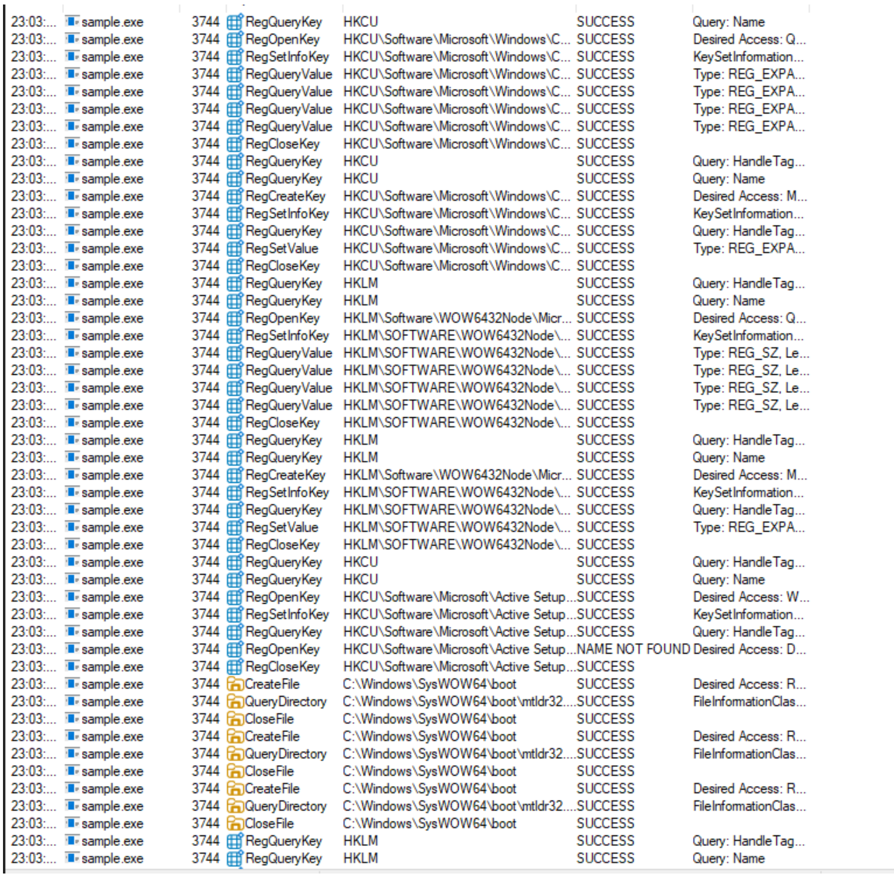
- `RegQueryKey`
- `RegOpenKey`
- `RegCreateKey`
- `RegSetValue`
- `RegSetInfoKey`

The sample also accessed registry locations such as
`HKCU\Software\Microsoft\Windows`, `HKLM\SOFTWARE`, `HKLM\SOFTWARE\WOW6432Node`, `HKCU\Software\Microsoft\Active Setup`

This behavior suggests:

- System environment discovery
- Operating system and configuration enumeration
- Potential persistence mechanism preparation

Notably, access to **Active Setup registry keys** indicates a possible attempt to establish persistence by executing malicious components during user login.



23:03:...	sample.exe	3744	RegQueryKey	HKCU	SUCCESS	Query: Name
23:03:...	sample.exe	3744	RegOpenKey	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Desired Access: Q...
23:03:...	sample.exe	3744	RegSetInfoKey	HKCU\Software\Microsoft\Windows\C...	SUCCESS	KeySetInformation...
23:03:...	sample.exe	3744	RegQueryValue	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Type: REG_EXPA...
23:03:...	sample.exe	3744	RegQueryValue	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Type: REG_EXPA...
23:03:...	sample.exe	3744	RegQueryValue	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Type: REG_EXPA...
23:03:...	sample.exe	3744	RegQueryValue	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Type: REG_EXPA...
23:03:...	sample.exe	3744	RegCloseKey	HKCU\Software\Microsoft\Windows\C...	SUCCESS	
23:03:...	sample.exe	3744	RegQueryKey	HKCU	SUCCESS	Query: HandleTag...
23:03:...	sample.exe	3744	RegQueryKey	HKCU	SUCCESS	Query: Name
23:03:...	sample.exe	3744	RegCreateKey	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Desired Access: M...
23:03:...	sample.exe	3744	RegSetInfoKey	HKCU\Software\Microsoft\Windows\C...	SUCCESS	KeySetInformation...
23:03:...	sample.exe	3744	RegQueryKey	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Query: HandleTag...
23:03:...	sample.exe	3744	RegSetValue	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Type: REG_EXPA...
23:03:...	sample.exe	3744	RegCloseKey	HKCU\Software\Microsoft\Windows\C...	SUCCESS	
23:03:...	sample.exe	3744	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
23:03:...	sample.exe	3744	RegQueryKey	HKLM	SUCCESS	Query: Name
23:03:...	sample.exe	3744	RegOpenKey	HKLM\Software\WOW6432Node\Micr...	SUCCESS	Desired Access: Q...
23:03:...	sample.exe	3744	RegSetInfoKey	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	KeySetInformation...
23:03:...	sample.exe	3744	RegQueryValue	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	Type: REG_SZ, Le...
23:03:...	sample.exe	3744	RegQueryValue	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	Type: REG_SZ, Le...
23:03:...	sample.exe	3744	RegQueryValue	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	Type: REG_SZ, Le...
23:03:...	sample.exe	3744	RegQueryValue	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	Type: REG_SZ, Le...
23:03:...	sample.exe	3744	RegCloseKey	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	
23:03:...	sample.exe	3744	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
23:03:...	sample.exe	3744	RegQueryKey	HKLM	SUCCESS	Query: Name
23:03:...	sample.exe	3744	RegCreateKey	HKLM\Software\WOW6432Node\Micr...	SUCCESS	Desired Access: M...
23:03:...	sample.exe	3744	RegSetInfoKey	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	KeySetInformation...
23:03:...	sample.exe	3744	RegQueryKey	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	Query: HandleTag...
23:03:...	sample.exe	3744	RegSetValue	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	Type: REG_EXPA...
23:03:...	sample.exe	3744	RegCloseKey	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	
23:03:...	sample.exe	3744	RegQueryKey	HKCU	SUCCESS	Query: HandleTag...
23:03:...	sample.exe	3744	RegQueryKey	HKCU	SUCCESS	Query: Name
23:03:...	sample.exe	3744	RegOpenKey	HKCU\Software\Microsoft\Active Setup...	SUCCESS	Desired Access: W...
23:03:...	sample.exe	3744	RegSetInfoKey	HKCU\Software\Microsoft\Active Setup...	SUCCESS	KeySetInformation...
23:03:...	sample.exe	3744	RegQueryKey	HKCU\Software\Microsoft\Active Setup...	SUCCESS	Query: HandleTag...
23:03:...	sample.exe	3744	RegOpenKey	HKCU\Software\Microsoft\Active Setup...	NAME NOT FOUND	Desired Access: D...
23:03:...	sample.exe	3744	RegCloseKey	HKCU\Software\Microsoft\Active Setup...	SUCCESS	
23:03:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:03:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:03:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:03:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:03:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:03:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:03:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:03:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:03:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:03:...	sample.exe	3744	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
23:03:...	sample.exe	3744	RegQueryKey	HKLM	SUCCESS	Query: Name

Fig 5.4.1 Actions performed by malware sample as well as Registry changes made

5.4.4 Loaded Libraries

During execution, the malware loaded multiple system libraries, including:

- **mscoree.dll** – Indicates execution within the Microsoft .NET runtime environment.
- **kernel32.dll** – Core Windows process and memory management functions.
- **advapi32.dll** – Registry and security-related operations.
- **winsock32.dll / wininet.dll** – Network communication capabilities.
- **combase.dll** – Component Object Model (COM) interaction.

The presence of .NET runtime libraries confirms that the malware is a managed .NET executable.The loading of networking libraries suggests potential command-and-control communication capability. But the most damning is the use of `advapi32.dll` which is used to communicate with the registry and make changes there, suggesting the capability of the malware to change the environment to maintain long term persistence within the system.

23:05:...	sample.exe	3744	RegQueryValue	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegQueryValue	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegQueryValue	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegQueryValue	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegCloseKey	HKCU\Software\Microsoft\Windows\C...	SUCCESS	
23:05:...	sample.exe	3744	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
23:05:...	sample.exe	3744	RegQueryKey	HKLM	SUCCESS	Query: Name
23:05:...	sample.exe	3744	RegOpenKey	HKLM\Software\WOW6432Node\Micr...	REPARSE	Desired Access: Q...
23:05:...	sample.exe	3744	RegOpenKey	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	Desired Access: Q...
23:05:...	sample.exe	3744	RegSetInfoKey	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	KeySetInformation...
23:05:...	sample.exe	3744	RegQueryValue	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegQueryValue	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegQueryValue	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegQueryValue	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegCloseKey	HKLM\SOFTWARE\Microsoft\Window...	SUCCESS	
23:05:...	sample.exe	3744	RegQueryKey	HKCU	SUCCESS	Query: HandleTag...
23:05:...	sample.exe	3744	RegQueryKey	HKCU	SUCCESS	Query: Name
23:05:...	sample.exe	3744	RegOpenKey	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Desired Access: Q...
23:05:...	sample.exe	3744	RegSetInfoKey	HKCU\Software\Microsoft\Windows\C...	SUCCESS	KeySetInformation...
23:05:...	sample.exe	3744	RegQueryValue	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegQueryValue	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegQueryValue	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegQueryValue	HKCU\Software\Microsoft\Windows\C...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegCloseKey	HKCU\Software\Microsoft\Windows\C...	SUCCESS	
23:05:...	sample.exe	3744	RegQueryKey	HKLM	SUCCESS	Query: HandleTag...
23:05:...	sample.exe	3744	RegQueryKey	HKLM	SUCCESS	Query: Name
23:05:...	sample.exe	3744	RegOpenKey	HKLM\Software\WOW6432Node\Micr...	SUCCESS	Desired Access: Q...
23:05:...	sample.exe	3744	RegSetInfoKey	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	KeySetInformation...
23:05:...	sample.exe	3744	RegQueryValue	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegQueryValue	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegQueryValue	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegQueryValue	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	Type: REG_EXPA...
23:05:...	sample.exe	3744	RegCloseKey	HKLM\SOFTWARE\WOW6432Node\...	SUCCESS	
23:05:...	sample.exe	3744	RegQueryKey	HKCU	SUCCESS	Query: HandleTag...
23:05:...	sample.exe	3744	RegQueryKey	HKCU	SUCCESS	Query: Name
23:05:...	sample.exe	3744	RegOpenKey	HKCU\Software\Microsoft\Active Setup...	SUCCESS	Desired Access: W...
23:05:...	sample.exe	3744	RegSetInfoKey	HKCU\Software\Microsoft\Active Setup...	SUCCESS	KeySetInformation...
23:05:...	sample.exe	3744	RegQueryKey	HKCU\Software\Microsoft\Active Setup...	SUCCESS	Query: HandleTag...
23:05:...	sample.exe	3744	RegOpenKey	HKCU\Software\Microsoft\Active Setup...	NAME NOT FOUND	Desired Access: D...
23:05:...	sample.exe	3744	RegCloseKey	HKCU\Software\Microsoft\Active Setup...	SUCCESS	
23:05:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:05:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:05:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:05:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:05:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:05:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:05:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:05:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:05:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	

Fig 5.4.2 Continued activity of malware sample

5.4.5 System Profiling Behavior

The malware kept performing process profiling tasks and system scans. This shows that the malware collects information about the target system before carrying out further malicious activities.

This kind of activity is typical of Remote Access Trojans that carry out system fingerprinting before setting up communication with a command and control server.
During the execution phase, the malware interacted with the following directory:

`C:\Windows\SysWOW64\boot`

The process searched for the presence of a file named mtldr32. The file system access was read-only, with no file writes detected. This activity indicates that the malware was checking the system settings rather than trying to persist at the boot level.

5.5 Behavioral Summary

From the runtime analysis, it can be concluded that the malware exhibits traits typical of a Remote Access Trojan. The malware carries out system reconnaissance, registry scanning, and possible persistence setup. It also loads networking libraries and keeps scanning system resources, which shows that it is preparing for remote control functionality.

There were no destructive activities or further payload execution during the analysis period, which shows that the malware might need external command and control communication to carry out further malicious activities.

Time ...	Process Name	PID	Operation	Path	Result	Detail
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:03:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:03:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:03:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:03:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:03:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:03:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:03:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:03:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:03:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:03:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:03:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:03:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:03:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:03:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:03:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:03:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:03:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:03:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:03:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:03:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:03:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:03:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:03:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:03:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	Process Profiling		SUCCESS	User Time: 0.0156...
23:03:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...
23:03:...	sample.exe	3744	QueryDirectory	C:\Windows\SysWOW64\boot\mtldr32....	SUCCESS	FileInformationClas...
23:03:...	sample.exe	3744	CloseFile	C:\Windows\SysWOW64\boot	SUCCESS	
23:03:...	sample.exe	3744	CreateFile	C:\Windows\SysWOW64\boot	SUCCESS	Desired Access: R...

Fig 5.4.3 Continued activity of malware sample

Registry Summary

Registry paths accessed during trace:

Registry Time	Total Events	Opens	Closes	Reads	Writes	Other	Path
4.4267293	7193	1086	816	2700	821	1770	<Total>
0.0000027	1	0	1	0	0	0	HKCR
0.0021333	814	0	0	0	0	814	HKCU
0.0003991	405	0	135	0	135	135	HKCU\Software\Microsoft\Active Set...
0.0010804	135	135	0	0	0	0	HKCU\Software\Microsoft\Active Set...
0.0009866	135	135	0	0	0	0	HKCU\Software\Microsoft\Active Set...
1.6538050	409	136	136	0	136	1	HKCU\Software\Microsoft\Windows\...
0.0009728	541	0	0	540	1	0	HKCU\Software\Microsoft\Windows\...
0.0040109	409	136	136	0	136	1	HKCU\Software\Microsoft\Windows\...
0.0019175	541	0	0	540	1	0	HKCU\Software\Microsoft\Windows\...
0.0054804	816	0	0	0	0	816	HKLM
0.0797494	1	0	0	0	1	0	HKLM\SOFTWARE\Microsoft\Wind...
0.0029306	409	136	136	0	136	1	HKLM\SOFTWARE\Microsoft\Wind...
0.0011695	541	0	0	540	1	0	HKLM\SOFTWARE\Microsoft\Wind...
0.0047208	409	136	136	0	136	1	HKLM\Software\WOW6432Node\Mi...
0.0010720	541	0	0	540	1	0	HKLM\SOFTWARE\WOW6432Nod...
0.3380074	136	136	0	0	0	0	HKLM\Software\WOW6432Node\Mi...
2.2920050	409	136	136	0	136	1	HKLM\Software\WOW6432Node\Mi...
0.0362859	541	0	0	540	1	0	HKLM\SOFTWARE\WOW6432Nod...

Fig 5.4.5: Registry Paths Accessed

Files accessed during trace:

By Path | By Folder | By Extension

Name	File Time	Total Events	Opens	Closes	Reads	Writes	Read B...	Write B...	Get ACL	Set ACL	Other
⊟ C:	5.9248855	1535	429	429	248	0	6342144	0	0	0	429
$Directory	0.0822957	9	0	0	9	0	36864	0	0	0	0
⊟ Users	0.0004839	1	0	0	1	0	512	0	0	0	0
⊟ mahes	0.0004839	1	0	0	1	0	512	0	0	0	0
⊟ Desktop	0.0004839	1	0	0	1	0	512	0	0	0	0
sample.exe	0.0004839	1	0	0	1	0	512	0	0	0	0
⊟ Windows	5.8421059	1525	429	429	238	0	6304768	0	0	0	429
⊟ SysWOW64	5.7889277	1468	429	429	181	0	5581312	0	0	0	429
AcGenral.dll	0.0044944	7	0	0	7	0	229376	0	0	0	0
AcLayers.dll	0.0014711	3	0	0	3	0	98304	0	0	0	0
KernelBase.dll	2.1528959	53	0	0	53	0	1637376	0	0	0	0
SHCore.dll	0.0073568	6	0	0	6	0	180224	0	0	0	0
advapi32.dll	0.0076700	6	0	0	6	0	196608	0	0	0	0
apphelp.dll	0.0065044	5	0	0	5	0	163840	0	0	0	0
⊟ boot	0.0938937	1287	429	429	0	0	0	0	0	0	429
mtldr32.exe	0.0000000	429	0	0	0	0	0	0	0	0	429
combase.dll	0.0076831	6	0	0	6	0	196608	0	0	0	0
kernel32.dll	0.0233264	22	0	0	22	0	696320	0	0	0	0
msacm32.dll	0.0007101	1	0	0	1	0	32768	0	0	0	0
mscoree.dll	0.0045176	6	0	0	6	0	196608	0	0	0	0
msctf.dll	0.0017702	2	0	0	2	0	65536	0	0	0	0
msvcrt.dll	0.1340849	1	0	0	1	0	32768	0	0	0	0
ntdll.dll	0.0645840	30	0	0	30	0	794624	0	0	0	0
propsys.dll	0.0138776	9	0	0	9	0	294912	0	0	0	0
shell32.dll	0.8240305	7	0	0	7	0	229376	0	0	0	0
ucrtbase.dll	0.0086759	5	0	0	5	0	163840	0	0	0	0
windows.storage.dll	2.4284280	7	0	0	7	0	229376	0	0	0	0
wininet.dll	0.0015436	2	0	0	2	0	65536	0	0	0	0
winspool.drv	0.0008423	2	0	0	2	0	65536	0	0	0	0
wsock32.dll	0.0005672	1	0	0	1	0	11776	0	0	0	0
⊟ System32	0.0531782	57	0	0	57	0	723456	0	0	0	0
ntdll.dll	0.0069177	7	0	0	7	0	110592	0	0	0	0
wow64.dll	0.0419211	45	0	0	45	0	610304	0	0	0	0
wow64cpu.dll	0.0043394	5	0	0	5	0	2560	0	0	0	0

Fig 5.4.6: Files Accessed

```
[DNS Query Received.]
 Domain name: 3.tlu.dl.delivery.mp.microsoft.com
[DNS Response sent.]

[DNS Query Received.]
 Domain name: 3.tlu.dl.delivery.mp.microsoft.com
[DNS Response sent.]

[DNS Query Received.]
 Domain name: 3.tlu.dl.delivery.mp.microsoft.com
[DNS Response sent.]

[DNS Query Received.]
 Domain name: 3.tlu.dl.delivery.mp.microsoft.com
[DNS Response sent.]

[DNS Query Received.]
 Domain name: 3.tlu.dl.delivery.mp.microsoft.com
[DNS Response sent.]

[DNS Query Received.]
 Domain name: 3.tlu.dl.delivery.mp.microsoft.com
[DNS Response sent.]

[DNS Query Received.]
 Domain name: 3.tlu.dl.delivery.mp.microsoft.com
[DNS Response sent.]

[DNS Query Received.]
 Domain name: self.events.data.microsoft.com
[DNS Response sent.]

[DNS Query Received.]
 Domain name: arc.msn.com
[DNS Response sent.]
[Received data over ICMP.]
 [ICMP Type: 3.]
 [ICMP Code: 1.]
 [ICMP Data:]
 E
[Received data over ICMP.]
 [ICMP Type: 3.]
 [ICMP Code: 1.]
 [ICMP Data:]
 E
[Received data over ICMP.]
 [ICMP Type: 3.]
 [ICMP Code: 1.]
 [ICMP Data:]
 E
```
Fig 5.4.7: Network connections made during execution

6. Conclusion

Analysis of the example Setup.exe reveals that it is a DaRCoMet Remote Access Trojan (RAT) intended for unauthorized access and control of the compromised computer, developed using the .NET framework and heavily integrated with Windows components.

During dynamic analysis, the malware is observed to perform registry enumeration and system reconnaissance for gathering host information and preparing the execution environment. It reads values from both HKCU and HKLM, including Active Setup directories, which indicates the potential for persistence mechanisms. It also performs system profiling and scans Windows directories, including the boot directory, which indicates the malware checks the environment and compatibility.

The malware sample dynamically loads various Windows libraries related to process manipulation, registry modification, security, and networking. The loading of libraries related to networking indicates the malware has the potential for command and control functionality, which is typical of RATs. However, there was no file system modification, payload execution, or destructive functionality observed during the analysis, which indicates that the malware sample was in its initial or reconnaissance stage and would require further commands from a remote C2 server for malicious functionality.

In conclusion, the malware sample analyzed has characteristics of a reconnaissance-oriented Remote Access Trojan, capable of system monitoring, preparing for persistence, and facilitating remote control, which indicates the potential danger if it establishes external connectivity.